

Portfolio Project - Stage 1 Report

Team Formation, Brainstorming and MVP

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1. Summary

This document defines the initial structure of the end-of-year project.

At this stage:

- The solo project approach is clearly defined and justified
- Personal organisation and working methods are established
- Key stakeholders are identified, along with how they will be engaged
- Several potential project directions have been identified, yet one option has been ultimately selected for the final MVP.

The project is intended to combine fullstack development, user experience, and a first approach to applied machine learning, in line with my future specialisation and personal interests.

2. Team Formation Overview

2.1. Solo project

This project will be carried out individually, this decision was made deliberately and is grounded in three practical reasons:

- **Full ownership of the project:** working alone allows me to choose a topic that is close to my interests, my values, and my long-term learning goals.
It also gives me full control over the technical and creative direction of the project, from the first idea to the final implementation.
- **Scope calibration:** another reason for this choice is that I want to experience every part of the development process myself (project planning, design, frontend, backend, testing, documentation, and possibly a light AI-related feature).
- **Accountability:** finally, working alone also helps avoid some common team-related issues, such as coordination problems, uneven workload, or unclear responsibilities.
This means that I will be fully responsible for the project's progress, decisions, strengths, and weaknesses.

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I am aware that working alone also brings risks - such as reduced feedback, isolation, or loss of perspective - and I will need to stay vigilant on these points; however, I choose to approach this constraint as an opportunity to grow, take full responsibility, and develop strong autonomy throughout the project.

Even though this is not a team project in the traditional sense, I will still apply clear working rules: organisation, role definition, regular progress reviews, and stakeholder identification.

2.2. Developer Profile

The following table summarises the background, strengths, and current skill level of the sole developer:

Attribute	Detail
Name	Felix BESANCON
Location	Bordeaux, France
Program	Bachelor CDA - Application Design & Development, Year 1
Background	Career switcher, with previous experience in optics, healthcare, and marketing
Core strengths	Structured thinking, problem solving, clean code practices, self-learning
Personal interests	Cinema, modern and tech-related art, creative coding, crafty hobbies (such as crochet and model making), and game experiences (video games, board games, role-playing games, and game mastering)
Career goal	Find a work-study placement from September/October 2026 and continue toward a fullstack / AI-oriented specialisation

2.3. Technical Role Definition

As this is a solo project, I will take on all the main roles needed during the project. This helps me understand the full development cycle and stay consistent in my choices:

Role	Responsibilities	Reason
Project Manager	Planning, task breakdown, timeline, risk tracking	Needed to keep the project realistic and organized
Fullstack Developer	Frontend, backend, API design, database logic	Main technical role, aligned with my future specialisation
UI/UX Designer	Wireframes, user flows, visual choices	Important for building a usable and clear product
Data / AI Developer	Simple data processing, recommendation logic, or AI-based feature	Linked to my fullstack / AI learning path, but kept realistic for Year 1
QA / Tester	Manual testing, bug tracking, self-review	Needed to maintain quality without a team
Technical Writer	README, documentation, project report	Important for clarity, maintainability, and evaluation

2.4. Communication and Collaboration Norms

In a solo project, communication rules become personal organization rules, personal working standards and self-accountability practices. The goal is to stay structured, avoid losing track of decisions, and keep a clear project history.

The following conventions are established for the duration of this project:

Area	Tool / Method	Purpose
Task management	GitHub Projects or Trello	Track tasks, backlog, and priorities
Documentation	Google Docs / Drive	Keep reports, notes, and decisions centralized
Version control	Git / GitHub	Save progress and keep a clean development history
Progress review	Weekly self-review	Check progress, blockers, and next priorities
Decision log	Dedicated document	Explain important technical and design choices

I will use clear commit messages, following a simple convention such as **'feat'**, **'fix'**, **'docs'**, and **'chore'**. This will make the Git history easier to read and understand.

3. Stakeholder Overview

Even as a solo project, several people or groups have an interest in the project's outcome, its quality, and its direction.

The following section identifies the main stakeholders and their role.

3.1. Stakeholder Register

Stakeholder	Type	Influence / Interest	Role & Impact
Holberton School instructors	Internal / Primary	High / High	Define evaluation criteria, review deliverables, and provide technical and methodological guidance throughout the project
Year-end jury / evaluators	External / Primary	High / High	Evaluate the final MVP (technical quality, clarity, innovation, presentation, relevance to the chosen specialisation)
Future employers / alternance recruiters	External / Primary	High / Medium	This project acts as a portfolio piece, its design, code quality, and documentation can directly influence hiring opportunities
End users	External / Secondary	Low / High	People who will actually use the application; their experience will guide usability User testing and feedback will shape design decisions
Holberton peer community	Internal / Secondary	Low / Medium	Can provide informal feedback, and code review by peers Potential source of user testing

3.2. Stakeholder Engagement Strategy

The following approach will be used to stay aligned with each stakeholder group:

- **Holberton instructors:**
Regular submissions and check-ins, with clear communication in case of blockers
 - **Jury / evaluators:**
A clear final presentation, including a working demo, documentation, and explanation of key choices
 - **Future employers:**
A clean and public GitHub repository, with readable commits, a clear README, and a demo if possible
 - **End users:**
Simple user testing once a prototype is ready, with feedback used to improve the product
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4. Brainstorming and Idea Evaluation

4.1. Personal context and creative starting point

Before listing the ideas themselves, it is useful to explain the mindset behind this brainstorming phase.

My interests go beyond technology: I am drawn to **art** (especially **cinema**, comics, and anything that combines **storytelling** with **visual content**), gamified or **playful experiences**, and tools that make creative things **more accessible**.

When thinking about a project, I naturally looked for ideas that were either **facilitating** (helping someone do something they couldn't before) or **fun** (creating an experience worth having).

One key criterion also emerged early in the process: regardless of how far development goes, **the final product should always be presentable**.

In other words, a stopped project at 60% should still produce a usable, demonstrable MVP - not just an unfinished skeleton.

This became one of the decisive selection factors.

4.2. Brainstorming session

Mind Mapping

Starting from my personal interests - cinema, storytelling, games, creative tools, applied AI - I mapped out four directions:

- **Film watching** → viewing history, taste profiling, recommendations, mood, streaming platforms, cinema
- **Creative world-building** → map generation, fantasy universes, storytelling tools, game masters
- **Music & rhythm playing** → interactive games, YouTube integration, beat detection, social sharing
- **Knowledge & community managing** → Wikipedia, accessible content creation, public interest tools

Each branch led to concrete ideas, which are described in detail below.

SCAMPER Framework

Reference tool: Letterboxd (popular film diary and social network for cinema lovers):

SCAMPER	Application
Substitute	Replace static star ratings with contextual tags and mood-based notation
Combine	Combine a viewing diary with an AI recommendation engine and streaming availability data
Adapt	Adapt the social discovery model to use mood as an input rather than just genre
Modify	Add a conversational AI layer to capture impressions in a more natural way
Put to another use	Use viewing history as training data for a personal taste profile

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SCAMPER	Application
Eliminate	Remove the need to search for where to watch - surface it automatically
Reverse	Instead of the user searching for a film, let the app suggest one based on how they feel right now

Reference tool: AI Fantasy Map Generator/getimg.ai/ (AI Map Generator for Game Masters & World Builders):

SCAMPER	Application
Substitute	Replace manual drawing tools with a colour-coded input system where each colour automatically maps to a terrain or element type
Combine	Combine a simplified drawing interface with an AI image generation layer to produce polished, detailed visuals from rough sketches
Adapt	Adapt the concept beyond game masters - apply it to authors, screenwriters, or children creating fantasy worlds without any drawing skills
Modify	Add a scale selector so the same tool can generate anything from a building interior to an entire continent
Put to another use	Use the generated maps as printable assets, shareable images, or exportable files for other creative tools
Eliminate	Remove the need for artistic skill entirely - the user only needs to communicate intent, not produce a finished drawing
Reverse	Instead of the user prompting and the AI drawing, let the user make the first sketch and the AI generate a map and then let the user adjust and annotate it to refine

Reference tool: Guitar Hero/Dance Dance Revolution/Friday Night Funkin'/Osu! (rhythm games):

SCAMPER	Application
Substitute	Replace a fixed song library with any YouTube video chosen by the user
Combine	Combine audio analysis, beat detection, and a browser-based game in one tool
Adapt	Adapt the rhythm game format to a web-first, no-install experience
Modify	Add difficulty settings and score sharing for a social layer
Put to another use	Use the generated levels as a social and streaming-friendly format - something people can share, compete on, or watch others play
Eliminate	Remove the need to own or download songs - use what already exists on YouTube
Reverse	Instead of playing songs the game decides, let the user bring their own music

"How Might We" Questions

- How might we help someone choose a film to watch tonight, based on how they actually feel right now?
 - How might we make a film diary more engaging than just filling in star ratings?
 - How might we let someone play a rhythm game on any song they already love, without installing anything?
 - How might we help someone with no drawing skills create a visual map for storytelling purposes?
 - How might we make contributing to a knowledge base as simple as filling in a form?
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5. Idea evaluation

5.1. Ideas explored

Idea 1 - Intelligent Film Viewing Journal (**selected**)

A **web application** where users **log** the films they watch and **rate** them.

The key feature is an **integrated AI tool** that makes **personalized recommendations** based on **viewing history**, other users with **similar tastes**, and the **user's current "mood"**.

The tool takes into account **current movie releases**, as well as the **streaming platforms** catalogs the users actually have access to.

The notation system would be **deliberately different** from standard star ratings, using **original tags and categories** to make the experience more engaging and worth the user's time.

Idea 2 - AI Map Generator for Game Masters and World Builders

A **web application** with an online **simplified drawing tool** (similar to a basic **Paint** application) where the user **sketches a map** using **colour-coded elements** (grey lines = stone walls, green hatching = grass, etc.).

The users could add **short text labels** to clarify their intent (e.g., "this is covered in parchment," "this is a staircase with one broken step," etc.), allowing for a **more precise sketch**.

An AI model then generates a **detailed, clean visual and playable map** ready to be used in tabletop role-playing games or creative writing.

The **scale could be adjusted** to generate anything from a room interior to an entire continent.

Idea 3 - YouTube Rhythm Game Generator

A **web application** where the user pastes a **YouTube URL**.

The tool's integrated **AI** analyses the audio track, detects the **beats and rhythm**, and generates a **playable rhythm game** level directly in the browser - displayed just below the **embedded YouTube player**.

The game would use **keyboard inputs** (tap and hold mechanics on four keys), with **adjustable difficulty**, **score tracking**, and the ability to **share generated levels** with others.

Idea 4 - Wikipedia Builder

A **simplified interface** to help users **create or enrich Wikipedia pages**, making the contribution process more **accessible** to people who find the current system too complex or technical.

5.2. Evaluation criteria

Criterion	Description
Feasibility	Can this be built by a solo Year 1 developer in approximately 3 months?
Technical alignment	Does it match the fullstack + light AI specialisation path?
MVP progressiveness	Is the project defensible even if development stops early?
Personal relevance	Is this a tool I would actually use and care about?
Market differentiation	Does it bring something that does not already exist in the same form?
Community fit	Is it aligned with the values or expectations of its target users?

5.3. Idea ranking

Rank	Idea	Feasibility	Technical alignment	MVP progress-iveness	Personal relevance	Differenti-ation	Commun-i ty fit
1	Intelligent Film Journal	High	Fullstack + AI API	Strong	Daily use	Original angle	Positive
2	Rhythm Game Generator	Medium	Complex ML + game	Risky	Very high	Unique	Positive
3	AI Map Generator	Low	Too AI-heavy	Risky	Indirect	Exists without AI	Negative
4	Wikipedia Builder	High	Light tech	Strong	Low	Contradict s the platform	Negative

Note: the Rhythm Game Generator scored highest on personal relevance, but was ranked second overall due to its technical complexity and risk of being unfinishable within the project timeframe. As it was deemed too ambitious for a first-year student, it could probably be pursued as a personal project.

6. Decision and Refinement

6.1. Selected MVP: Intelligent Film Viewing Journal

The problem it solves:

Many people - myself included - regularly face the same situation: you want to watch something tonight, **but you don't know what** and before you know it, you end up spending the evening scrolling through your streaming service's catalog instead of enjoying a good movie as planned.

One night, you might feel like watching an award-winning masterpiece, just as you're entitled to defend bad taste the next day; it's totally fine to want to watch a B-movie with friends, or to look for a movie to watch while scrolling through your phone.

Existing tools like Letterboxd, IMDb, or AlloCiné let you rate films and browse catalogues, but they rarely combine your personal taste profile, your current mood, and the reality of what is actually available on your streaming subscriptions or at nearby theaters.

Beyond recommendations, there is also the question of memory: people watch a lot of films and rarely keep track of what they've seen, what they thought of it, and why. Standard star ratings feel reductive - a film can be considered as "mediocre" but also a "**comfort watch**", a "**guilty pleasure**", or "**perfect to put on in the background**", without being technically great or terrible.

The solution:

A web application that acts as a **personal film diary** and **intelligent recommendation engine**. The user logs in, records the films they watch, and rates them using an original and disruptive tagging system - more expressive and fun than simple star ratings. Over time, the app builds a taste profile and uses it, combined with mood input and streaming platform data, to suggest what to watch next.

Target audience:

- Cinema lovers and regular streaming users who feel overwhelmed by choice
 - Groups of people who have trouble agreeing on which movie to watch
 - People who like to keep track of what they have watched
 - Users looking for recommendations that go beyond "because you watched X, try Y"
 - Primarily desktop/web users, with a desire to adopt a mobile-first evolution later
-

Type of application:

- A responsive web application, accessible via browser on desktop and mobile.
 - Built with:
 - React (frontend)
 - Python with FastAPI (backend)
 - relational database (PostgreSQL or MySQL)
 - AI recommendations will be powered via an external LLM API (such as OpenAI or Mistral), combined possibly with the TMDB API for film data and the Watchmode API for streaming availability.
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Why this idea over the others

- It is the most balanced option across all evaluation criteria: feasible, technically aligned, personally meaningful, and with a clear differentiator (mood + streaming + original tagging)
- The project is progressive by design: even a basic version (log films, view history, see basic info via TMDB) is already a usable and demonstrable product - which means development can be paused at any stage without losing the ability to present something
- The AI layer is real but realistic: using an external LLM API or a simple recommendation algorithm avoids the need to train a custom model, which is beyond the scope of a Year 1 project
- When compared to other projects, this one offers unparalleled advantages for portfolio project:
 - Unlike the Map Generator, it does not rely on a community that is hostile to AI

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- Unlike the Wikipedia Builder, it does not work against the philosophy of its subject matter
 - Unlike the Rhythm Game, it does not risk becoming unfinishable
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Key features and SMART objectives:

Feature 1 - Film logging and viewing history

Goal: *By the end of Month 1, a logged-in user can search for any film via the TMDB API, add it to their viewing history, and retrieve that history on a personal dashboard.*

This is the foundation of the entire product. Without it, nothing else works.

Feature 2 - Original tagging and rating system

Goal: *By the end of Month 2, users can rate a film using at least 8 custom tags in addition to (or instead of) a standard score. Tags should include contextual and emotional categories.*

Examples of tags being considered:

- 🎉 Great with a group
- 📺 Perfect background watch
- 😭 Emotional wreck (in a good way)
- 😊 Guilty pleasure
- 🤯 Mind blowing
- 📱 Can scroll while watching
- 🤡 So stupid it's good
- 🧠 Needs full attention
- 💤 Fell asleep (not necessarily a bad thing)
- 🔄 Would rewatch immediately
- 🎬 Instant classic

Feature 3 - Mood-based AI recommendation

Goal: *By the end of Month 3, the app can accept a mood input from the user and return at least 3 personalized film recommendations, filtered by the streaming platforms the user has selected in their profile, using LLM-based logic combined with their viewing history.*

Scope

In-scope (MVP)

- User authentication (registration and login)
- Film search and logging via TMDB API
- Personal viewing history and dashboard
- Custom tagging and rating system
- Basic user profile with streaming platform preferences
- Mood-based film recommendation (via LLM API + TMDB)
- Streaming availability display (via Watchmode API or equivalent)

Out-of-scope (post-MVP / future versions)

- Native mobile application
 - Social features (following other users, shared lists, comments)
 - Collaborative filtering based on a large user base (requires significant user volume)
 - Recommendation model trained on proprietary data
 - Integration with cinema theatres, or TV channel listings and showtimes (too dependent on third-party data availability)
 - TV series and episode tracking (possible extension, kept out of initial scope)
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Identified risks and mitigation strategies

Risk	Likelihood	Impact	Mitigation
Underestimation of the overall complexity of the project	Low	High	Break the project down into small, independently deliverable milestones. Prioritize core features strictly. If a feature proves too complex, document it as a future improvement rather than blocking progress.

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Risk	Likelihood	Impact	Mitigation
			Schedule a mid-project review with an instructor to reassess scope if needed.
Watchmode API limitations (rate limits, catalogue gaps)	Medium	Medium	Identify a backup API early (JustWatch unofficial, or omit streaming filter for MVP if needed)
LLM API costs becoming too high during development	Low	Medium	Use free tiers (Mistral, OpenAI free quota) and cache responses where possible
Project scope creeping beyond Year 1 capacity	High	High	Stick to the defined in-scope list; treat out-of-scope items as a documented future roadmap
Low user engagement during testing (no community yet)	Medium	Low	Focus on solo usability first; recruit 3–5 testers from the Holberton peer community
Overlapping with a peer's similar project	Medium	Low	Differentiation is clear: AI recommendation + mood + streaming integration is the unique angle

7. Documentation Summary

7.1. Team formation

This project is carried out solo by **Félix Besançon**, a Year 1 student at Holberton School Bordeaux, preparing for a fullstack in Year 2 and a machine learning specialisation after that.

All technical roles (Project Manager, Fullstack Developer, UI/UX Designer, Data/AI Developer, QA, Technical Writer) are handled by the same person.

Working tools include GitHub Projects and Trello for task management, Google Drive for documentation, and Git for version control with conventional commit messages.

Stakeholders include Holberton instructors, the year-end jury, future employers and alternance recruiters, and end users.

7.2. Selected MVP summary - CinéMood

The brainstorming phase explored four directions: a **smart film journal**, an **AI map generator for game masters**, a **YouTube rhythm game generator**, and a **Wikipedia contribution tool**.

All ideas were evaluated against six criteria (feasibility, technical alignment, MVP progressiveness, personal relevance, differentiation, and community fit) using structured techniques including Mind Mapping, the SCAMPER framework, and "How Might We" questions.

CinéMood, the smart film journal, emerged as the most balanced option and was selected as the final MVP.

Portfolio Project - Selected MVP:

Project name:

- **CinéMood** (*working title*)

What it is:

- A responsive web application for logging films, rating them with original tags, and receiving AI-powered recommendations based on viewing history, overall mood, and streaming platform availability.

The problem it solves:

- Choosing what to watch is harder than it should be. Existing tools either track films without recommending them intelligently, or recommend without knowing what you actually have access to or how you feel right now.

The solution it proposes:

- A personal, intelligent film companion that knows your taste, understands your mood, and only suggests films you can actually watch tonight.

Why it was selected:

- It is the most balanced idea across all criteria. It is buildable by a solo Year 1 developer, personally meaningful, and defensible at any stage of development. It also represents a genuine use case - a tool the developer would use every day.

Potential impact:

- Even at MVP level, this is a complete and usable product. As it grows, it could become a genuinely useful tool for any cinema lover who has ever stared at a streaming homepage and felt paralysed by choice.